

Math 6108 Homework 10

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Question 1.

Suppose that $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, and $B, C \in \mathcal{M}_{mn}(\mathbb{F})$. Then $B^*B = C^*C$ if and only if there exists a unitary matrix $U \in \mathcal{M}_m(\mathbb{F})$ such that $B = UC$.

Proof. If there is a unitary matrix $U \in \mathcal{M}_m(\mathbb{F})$ such that $B = UC$, then certainly

$$B^*B = C^*U^*UC = C^*C.$$

Conversely, suppose that $B^*B = C^*C$. Let $K = \{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq \mathbb{F}^m$ be an orthonormal basis for $C(\mathbb{R}^n)$, which we can construct by using the Gram-Schmidt process. Basis extension plus further Gram-Schmidt process implies that we can extend this basis to an orthonormal basis $M = \{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_m\} \subseteq \mathbb{F}^m$ that spans $\mathbb{F}^m$.

We can choose $L = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \subseteq \mathbb{F}^n$ such that $C\mathbf{u}_i = \mathbf{v}_i$. If L were linearly dependent, then $K = C(L)$ would also be linearly dependent; hence, L is linearly independent.

Claim: For all $\mathbf{y} \in \mathbb{F}^m$, there exists $\mathbf{u} \in \text{span}(L)$ and $\mathbf{h} \in \text{Null}(C)$ such that $\mathbf{y} = \mathbf{u} + \mathbf{h}$. Since $C\mathbf{u} \in \text{span}(K)$ by construction, it follows that there exists $a_1, \dots, a_k \in \mathbb{F}$ such that

$$C\mathbf{y} = a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k.$$

Choose $\mathbf{u} = a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k$, and choose $\mathbf{h} = \mathbf{y} - \mathbf{u}$. Then $\mathbf{u} \in \text{span}(L)$, and $\mathbf{y} = \mathbf{u} + \mathbf{h}$, and

$$\begin{aligned} C\mathbf{h} &= C\mathbf{y} - C\mathbf{u} = a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k - C(a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k) \\ &= a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k - (a_1\mathbf{v}_1 + \dots + a_k\mathbf{v}_k) = \mathbf{0}, \end{aligned}$$

so $\mathbf{h} \in \text{Null}(C)$.

Observe that $\text{Null}(B) = \text{Null}(C)$ because for any $\mathbf{x} \in \mathbb{R}^n$,

$$B\mathbf{x} = \mathbf{0} \iff \|B\mathbf{x}\|^2 = 0 \iff \mathbf{x}^*B^*B\mathbf{x} = 0 \iff \mathbf{x}^*C^*C\mathbf{x} = 0 \iff \|C\mathbf{x}\|^2 = 0 \iff C\mathbf{x} = \mathbf{0}.$$

This implies that the rank of B is the same as the rank of C by the rank-nullity theorem. The rank of C is k , so the rank of B is also k , which is also the rank of B^* . Then $\text{Nullity}(B^*) = m - k$ because $B^* \in \mathcal{M}_{nm}(\mathbb{F})$. Let $N = \{\mathbf{n}_1, \dots, \mathbf{n}_{m-k}\} \subseteq \mathbb{F}^m$ be an orthonormal basis for the null space of B^* .

Now define the matrix $U \in \mathcal{M}_m(\mathbb{F})$ by

$$U = \begin{bmatrix} B\mathbf{u}_1 & \dots & B\mathbf{u}_k & \mathbf{n}_1 & \dots & \mathbf{n}_{m-k} \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^* \\ \vdots \\ \mathbf{v}_k^* \\ \mathbf{v}_{k+1}^* \\ \vdots \\ \mathbf{v}_m^* \end{bmatrix}.$$

Then $B = UC$. To prove this, let $\mathbf{y} \in \mathbb{R}^n$ be given. Choose $\mathbf{u} \in \text{span}(L)$ and $\mathbf{h} \in \text{Null}(C) = \text{Null}(B)$ such that $\mathbf{y} = \mathbf{u} + \mathbf{h}$. Then

$$\mathbf{u} = a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k,$$

so

$$C\mathbf{y} = a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k.$$

Since M is orthonormal, it follows that $a_i = \mathbf{v}_i^* C\mathbf{y}$ for $i = 1, 2, \dots, m$, and $a_i = 0$ if $i > k$. This implies that

$$\begin{bmatrix} \mathbf{v}_1^* \\ \vdots \\ \mathbf{v}_k^* \\ \mathbf{v}_{k+1}^* \\ \vdots \\ \mathbf{v}_m^* \end{bmatrix} C\mathbf{y} = \begin{bmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Thus,

$$UC\mathbf{y} = \begin{bmatrix} B\mathbf{u}_1 & \cdots & B\mathbf{u}_k & \mathbf{n}_1 & \cdots & \mathbf{n}_{m-k} \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_k \\ 0 \\ \vdots \\ 0 \end{bmatrix} = a_1 B\mathbf{u}_1 + \cdots + a_k B\mathbf{u}_k = B(a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k) = B\mathbf{u}.$$

Since $\mathbf{h} \in \text{Null}(B)$, it follows that

$$UC\mathbf{y} = B\mathbf{u} = B(\mathbf{u} + \mathbf{h}) = B\mathbf{y}.$$

The vector $\mathbf{y} \in \mathbb{F}^n$ was arbitrary, so $B = UC$. Lastly, we observe that U is unitary. Let $\mathbf{x} \in \mathbb{F}^m$. There exists $a_1, \dots, a_m$ such that

$$\mathbf{x} = a_1 \mathbf{v}_1 + \cdots + a_m \mathbf{v}_m$$

because M is a basis for $\mathbb{F}^m$. Then by the orthonormality of M ,

$$U\mathbf{x} = \begin{bmatrix} B\mathbf{u}_1 & \cdots & B\mathbf{u}_k & \mathbf{n}_1 & \cdots & \mathbf{n}_{m-k} \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} = B(a_1 \mathbf{u}_1 + \cdots + a_k \mathbf{u}_k) + a_{k+1} \mathbf{n}_1 + \cdots + a_m \mathbf{n}_{m-k}.$$

Thus,

$$\begin{aligned} \begin{bmatrix} \mathbf{u}_1^* B^* \\ \vdots \\ \mathbf{u}_k^* B^* \\ \mathbf{n}_1^* \\ \vdots \\ \mathbf{n}_{m-k}^* \end{bmatrix} U\mathbf{x} &= \begin{bmatrix} a_1 \mathbf{u}_1^* B^* B\mathbf{u}_1 + \cdots + a_k \mathbf{u}_1^* B^* B\mathbf{u}_k + a_{k+1} \mathbf{u}_1^* B^* \mathbf{n}_1 + \cdots + a_m \mathbf{u}_1^* B^* \mathbf{n}_{m-k} \\ \vdots \\ a_1 \mathbf{u}_k^* B^* B\mathbf{u}_1 + \cdots + a_k \mathbf{u}_k^* B^* B\mathbf{u}_k + a_{k+1} \mathbf{u}_k^* B^* \mathbf{n}_1 + \cdots + a_m \mathbf{u}_k^* B^* \mathbf{n}_{m-k} \\ a_1 \mathbf{n}_1^* B\mathbf{u}_1 + \cdots + a_k \mathbf{n}_1^* B\mathbf{u}_k + a_{k+1} \mathbf{n}_1^* \mathbf{n}_1 + \cdots + a_m \mathbf{n}_1^* \mathbf{n}_{m-k} \\ \vdots \\ a_1 \mathbf{n}_{m-k}^* B\mathbf{u}_1 + \cdots + a_k \mathbf{n}_{m-k}^* B\mathbf{u}_k + a_{k+1} \mathbf{n}_{m-k}^* \mathbf{n}_1 + \cdots + a_m \mathbf{n}_{m-k}^* \mathbf{n}_{m-k} \end{bmatrix} \\ &= \begin{bmatrix} a_1 \mathbf{u}_1^* C^* C\mathbf{u}_1 + \cdots + a_k \mathbf{u}_1^* C^* C\mathbf{u}_k \\ \vdots \\ a_1 \mathbf{u}_k^* C^* C\mathbf{u}_1 + \cdots + a_k \mathbf{u}_k^* C^* C\mathbf{u}_k \\ a_{k+1} \\ \vdots \\ a_m \end{bmatrix} \end{aligned}$$

because $\mathbf{n}_i \in \text{Null}(B^*)$ for $i = 1, \dots, m - k$, and $B^*B = C^*C$, and by using the orthonormality of N . Since $\mathbf{v}_i = C\mathbf{u}_i$ for $i = 1, \dots, k$, and because M is orthonormal, we get

$$\begin{bmatrix} \mathbf{u}_1^* B^* \\ \vdots \\ \mathbf{u}_k^* B^* \\ \mathbf{n}_1^* \\ \vdots \\ \mathbf{n}_{m-k}^* \end{bmatrix} U \mathbf{x} = \begin{bmatrix} a_1 \\ \vdots \\ a_k \\ a_{k+1} \\ \vdots \\ a_m \end{bmatrix}.$$

Finally, this implies that

$$U^* U \mathbf{x} = \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_k & \mathbf{v}_{k+1} & \cdots & \mathbf{v}_m \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_k \\ a_{k+1} \\ \vdots \\ a_m \end{bmatrix} = a_1 \mathbf{v}_1 + \dots + a_m \mathbf{v}_m = \mathbf{x}.$$

Since $\mathbf{x} \in \mathbb{F}^m$ was arbitrary, it follows that $U^*U = I$, which implies that U is unitary. $\square$

Question 2.

Let $\lambda \in \mathbb{C}$ be an eigenvalue of a matrix $A \in \mathcal{M}_n(\mathbb{C})$. Then $|\lambda| \leq \|A\|$.

Proof. Let $\mathbf{v} \in \mathbb{C}^n$ be a unit eigenvector corresponding to λ . Then

$$\|A\| = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| \geq \|A\mathbf{v}\| = \|\lambda\mathbf{v}\| = |\lambda|.$$

$\square$

Question 3.

Let $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{n,p}$.

(a) Let $B = [\mathbf{b}_1, \dots, \mathbf{b}_p]$, where $\mathbf{b}_i \in \mathbb{C}^n$ for $i = 1, \dots, p$. Then $AB = [A\mathbf{b}_1, \dots, A\mathbf{b}_p]$, and

$$\|AB\|_F^2 = \sum_{i=1}^p \|A\mathbf{b}_i\|^2 \leq \|A\|^2 \sum_{i=1}^p \|\mathbf{b}_i\|^2 \leq \|A\|^2 \|B\|_F^2.$$

Hence, $\|AB\|_F \leq \|A\| \|B\|_F$.

(b) Let $\mathbf{x} \in \mathbb{C}^n$ be a unit vector, and let $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{C}^n$ be the rows of A . Then, by the Cauchy-Schwarz inequality,

$$\|A\mathbf{x}\|^2 = \sum_{i=1}^m |\mathbf{a}_i \mathbf{x}|^2 = \sum_{i=1}^m |\langle \mathbf{a}_i^*, \mathbf{x} \rangle|^2 \leq \sum_{i=1}^m \|\mathbf{a}_i^*\|^2 \|\mathbf{x}\|^2 = \sum_{i=1}^m \|\mathbf{a}_i\|^2 = \|A\|_F^2.$$

Thus, $\|A\mathbf{x}\| \leq \|A\|_F$ for all unit vectors $\mathbf{x}$. We conclude that

$$\|A\| = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| \leq \|A\|_F.$$

From part (a), it follows that

$$\|AB\|_F \leq \|A\|_F \|B\|_F.$$

Question 4.

To show that the operator norm $\|\cdot\|$ on $\mathcal{M}_{m,n}(\mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, is indeed a norm, we need to prove

1. For all $A \in \mathcal{M}_{m,n}(\mathbb{F})$, we have $\|A\| \geq 0$, with equality if and only if $A = 0$.
2. For all $A \in \mathcal{M}_{m,n}(\mathbb{F})$ and all $\lambda \in \mathbb{F}$, we have $\|\lambda A\| = |\lambda| \|A\|$.
3. For all $A, B \in \mathcal{M}_{m,n}(\mathbb{F})$, we have $\|A + B\| \leq \|A\| + \|B\|$.

To prove 1., we let $A \in \mathcal{M}_{m,n}(\mathbb{F})$. Choose any unit vector $\mathbf{x} \in \mathbb{F}^n$. Then $\|A\mathbf{x}\| \geq 0$ implies that $\|A\| \geq 0$. Now suppose that $A = 0$. Then $\|A\mathbf{x}\| = 0$ for all $\mathbf{x} \in \mathbb{F}^n$, so $\|A\| = 0$. On the other hand, if $\|A\| = 0$, then $0 \leq \|A\mathbf{x}\| \leq \|A\| = 0$ for any unit vector $\mathbf{x} \in \mathbb{F}^n$. Then for any $\mathbf{v} \in \mathbb{F}^n$, we can write $\mathbf{v} = \alpha \mathbf{x}$, where $\alpha \in \mathbb{F}$ and $\mathbf{x}$ is a unit vector. Then $\|A\mathbf{v}\| = |\alpha| \|A\mathbf{x}\| = 0$, which implies that $A\mathbf{v} = 0$ for all vectors $\mathbf{v} \in \mathbb{F}^n$. This means that $A = 0$.

To prove 2., we let $A \in \mathcal{M}_{m,n}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then

$$\|\lambda A\| = \max_{\|\mathbf{x}\|=1} \|\lambda A\mathbf{x}\| = \max_{\|\mathbf{x}\|=1} |\lambda| \|A\mathbf{x}\| = |\lambda| \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| = |\lambda| \|A\|.$$

To prove 3., we let $A, B \in \mathcal{M}_{m,n}(\mathbb{F})$. Then

$$\|A + B\| = \max_{\|\mathbf{x}\|=1} \|(A + B)\mathbf{x}\| \leq \max_{\|\mathbf{x}\|=1} (\|A\mathbf{x}\| + \|B\mathbf{x}\|) \leq \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\| + \max_{\|\mathbf{x}\|=1} \|B\mathbf{x}\| = \|A\| + \|B\|.$$

Question 5.

Suppose that $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, and $A \in \mathcal{M}_{m,n}(\mathbb{F})$. If the rank of A is r and A has singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, then

$$\|A\| = \sigma_1, \quad \|A\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2}.$$

Proof. Let $A = UDV^*$ be a singular value decomposition of A . Suppose that $\mathbf{v}_1$ is the first column of V . Then

$$\|A\| \geq \|A\mathbf{v}_1\| = \|UDV^*\mathbf{v}_1\| = \|D\mathbf{e}_1\| = \sigma_1,$$

where $\mathbf{e}_1 = [1 \ 0 \ \dots \ 0]^T \in \mathbb{F}^n$. On the other hand, if $\mathbf{x} \in \mathbb{F}^n$ is a unit vector, then so is $\mathbf{y} = V^*\mathbf{x}$ because V^* is unitary, and

$$\|A\mathbf{x}\| = \|UDV^*\mathbf{x}\| = \|D\mathbf{y}\| = \sqrt{\sum_{i=1}^r |\sigma_i y_i|^2} \leq \sqrt{\sum_{i=1}^r \sigma_i^2 |y_i|^2} \leq \sigma_1 \|\mathbf{y}\| = \sigma_1,$$

where $\mathbf{y} = [y_1 \ \cdots \ y_n]^T$. Taking the maximum over all unit vector $\mathbf{x} \in \mathbb{F}^n$, we get $\|A\| \leq \sigma_1$. Thus, $\|A\| = \sigma_1$.

Furthermore,

$$\|A\|_F = \text{Tr}(A^*A) = \text{Tr}(VD^*U^*UDV^*) = \text{Tr}(V^*VD^*D) = \text{Tr}(D^*D) = \sum_{i=1}^r \sigma_i^2.$$

Thus,

$$\|A\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2}.$$

□

Question 6.

Let

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}.$$

(a) The rank of A is 1 because its rows are clearly linearly dependent, implying its rank is less than 2, but $A \neq 0$, implying its rank is greater than 0. Then $\|A\| = \sigma_1 = \|A\|_F$ by Question 5. Hence, $\|A\| = \sqrt{1+1+1+1} = 2$.

(b) The characteristic polynomial of A is

$$P_A(\lambda) = \det \begin{bmatrix} 1-\lambda & -1 \\ -1 & -1-\lambda \end{bmatrix} = -(\lambda-1)(\lambda+1) + 1 = \lambda^2.$$

Thus, 0 is the only eigenvalue of A , and it has multiplicity 2. Since the Jordan canonical form J of A is 2×2 , the only two possibilities are two Jordan blocks of size 1 or one Jordan block of size 2. If there were two Jordan blocks of size 1, then we would have $J = 0$, which implies $A = 0$, which is false. Hence, there is one Jordan block of size 2; that is,

$$J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

(c) Note that

$$A^2 = 0,$$

so $A^n = 0$ for all $n \geq 2$. Then

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!} = I + A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}.$$